

Came up with an idea. Believe this should be a tidy fix to both performance and permissions issues with echoview idx files. The idea is we use the rclone '[union](#)' feature to direct all writes to a local dir instead of back to gcs and leave GCS read only. That means you can actually limit users who need to use echoview, but shouldn't be able to write to GCS, to GCS read permissions, and echoview should still work if set up this way. Steps to implement solution:

1. Create a file on your pam_user Desktop called rclone_custom.conf, and copy the below contents to the file:

```
[active-acoustics-gcs]
type = google cloud storage
project_number = ggn-nmfs-aa-dev-1
bucket_policy_only = true
env_auth = true

[echoview-union]
type = union
upstreams = C:\Users\pam_user\AppData\Local\Echoview_Workspace
active-acoustics-gcs:ggn-nmfs-aa-dev-1-data:ro
```

2. Fire off the custom rclone mount in a command line:

```
C:\Windows\rclone-v1.68.1-windows-amd64\rclone mount echoview-union: F: -o
UserName=pam_user --vfs-cache-mode full --vfs-cache-max-size 50G --file-perms 0777
--dir-perms 0777 --network-mode --config=C:\Users\pam_user\Desktop\rclone_custom.conf
```

Expected behaviors (should test):

- Path Echoview to F drive and use data normally. You should see fast performance around idx reads, perhaps even better than over network on-prem (ALL writes become cached on local).
- The GCS bucket is only available as read only. No action locally will affect the data on GCS.

Caveats:

- your local drive will gradually fill up with idx files unless these are super super tiny and doesn't matter if they accumulate locally. We can engineer something simple in the background to manage this, if it's a concern.
- Certain actions from the F drive, like deletes, seem to get a little funky locally with the union mount trying to figure out what's going on. Given the particularities of this workflow, likely a non-issue. Just use echoview normally, and don't use this dir for uses other than echoview for best behaviors.

BONUS: since the bucket is read only, feasible to have the mount fully baked into the acoustics windows workstation at startup for end users (users may not even need to perform gcloud login step, we could just permission workstation service account for read).